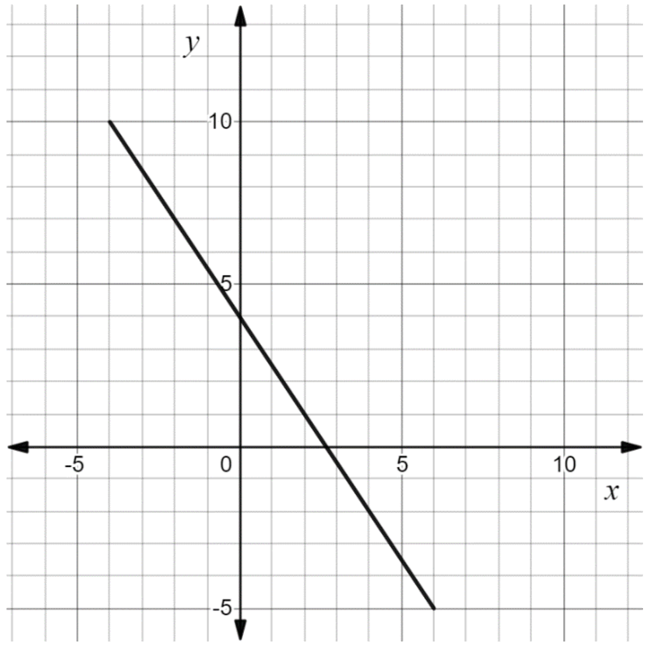
Algebra 1
Unit 4
Lesson 1 Homework

Name Answer Key____________
Date___________________
Period__________________

The graph of $y = -\frac{3}{2}x + 4$ is shown. The graph has a restricted domain.



1. Give a written description for the domain.
   All values greater than or equal to -4 and less than or equal to 6.

2. Write the range using inequalities.
   $-5 \leq y \leq 10$

3. Write the domain using set-builder notation.
   $\{x: -4 \leq x \leq 6\}$

4. Write the range using set-builder notation.
   $\{y: -5 \leq y \leq 10\}$

For each written description complete the table.

Written Description	Inequalities	Set-builder notation
5. All values less than or equal to -4	$x \leq -4$	$\{x: x \leq -4\}$
6. All real numbers	$-\infty < x < \infty$	$\{x: x \in \mathbb{R}\}$
7. All values greater than $-\frac{11}{3}$ but less than or equal to $\frac{17}{3}$	$-\frac{11}{3} < x \leq \frac{17}{3}$	$\left\{x: -\frac{11}{3} < x \leq \frac{17}{3}\right\}$

8. Write the set of values represented by $-6.7 \leq x < 9.1$ using set-builder notation.
$\{x: -6.7 \leq x \leq 9.1\}$

9. Write a written description of the set of values represented by $\{y \mid y \geq -0.25\}$.
All values greater than or equal to -0.25.

10. Write the set of values represented by $-14.25 \leq b \leq -8.37$ using set-builder notation.
$\{b: -14.25 \leq b \leq -8.37\}$

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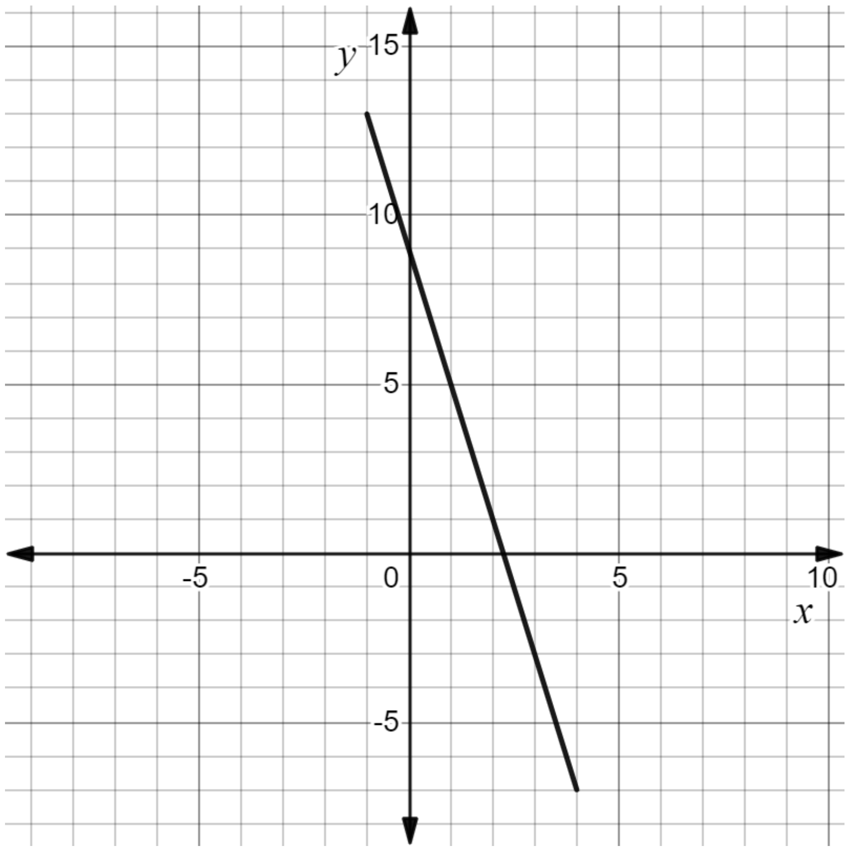
Lesson 2 Homework Period__________________

The graph of linear function is shown. The graph has a restricted domain.



1. The slope of the line segment is $-\frac{4}{1} = -4$.

2. As the x-values are increasing, the y-values are decreasing.

3. The coordinates of the point where the line crosses the y-axis are $(0,9)$.
The y-value of the point is 9.

4. The coordinates of the point where the line crosses the x-axis are $(2.25,0)$.
The x-value of the point is 2.25.

Identify the key features of the horizontal line $y = -6$ in questions 5-6.

5.

Domain	Range	As x-values increase, y-values ...
$\{x: -\infty < x < \infty\}$	$\{y: y = -6\}$	stays the same

6.

Slope	x-intercept	y-intercept
0	none	$(0, -6)$

Identify the key features of the line $y = \frac{6}{5}x + 11$ in questions 7-8.

7.

Domain	Range	As x-values increase, y-values ...
$\{x: -\infty < x < \infty\}$	$\{y: -\infty < y < \infty\}$	increases

8.

Slope	x-intercept	y-intercept
$\frac{6}{5}$	$\left(-\frac{55}{6}, 0\right)$	$(0,11)$

Identify the key features of the line $y = -\frac{5}{3}x - 2\frac{3}{7}$ in questions 9-10.

9.

Domain	Range	As x-values increase, y-values ...
$\{x: -\infty < x < \infty\}$	$\{y: -\infty < y < \infty\}$	decreases

10.

Slope	x-intercept	y-intercept
$-\frac{5}{3}$	$\left(-\frac{51}{35}, 0\right)$	$\left(0, -2\frac{3}{7}\right)$

Algebra 1
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Lesson 3 Homework

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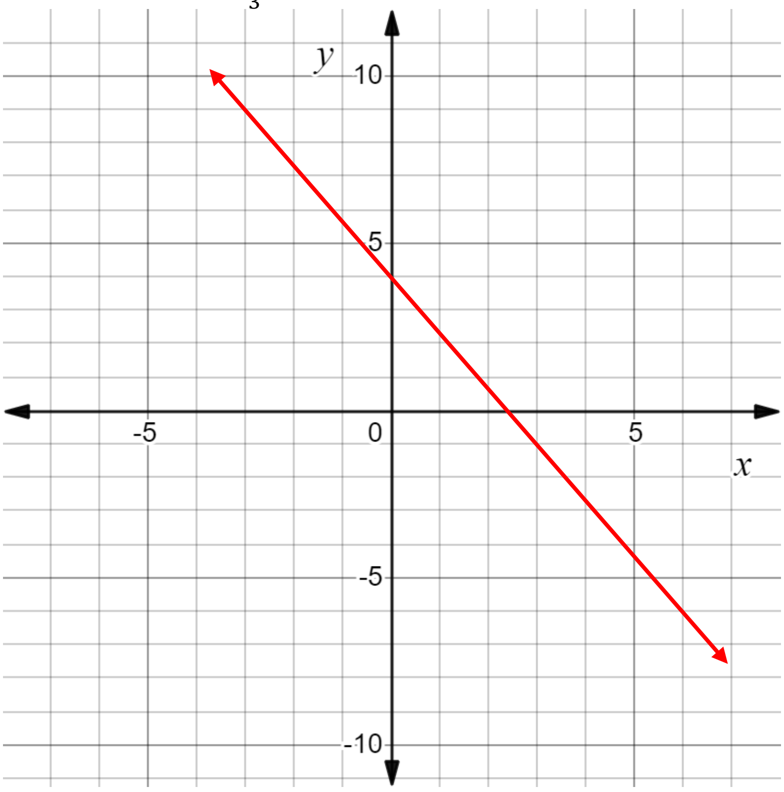
1. Complete the table of values for $y = -\frac{5}{3}x + 4$.

$\boldsymbol{x}$	-9	-6	0	3	6
$\boldsymbol{y}$	19	14	4	-1	-6

2. Graph the equation $y = -\frac{5}{3}x + 4$.



3. Write the domain of the line $y = -\frac{5}{3}x + 4$ using inequalities.
$-\infty < x < \infty$

4. What is the y-intercept of the line $y = -\frac{5}{3}x + 4$?
$(0,4)$

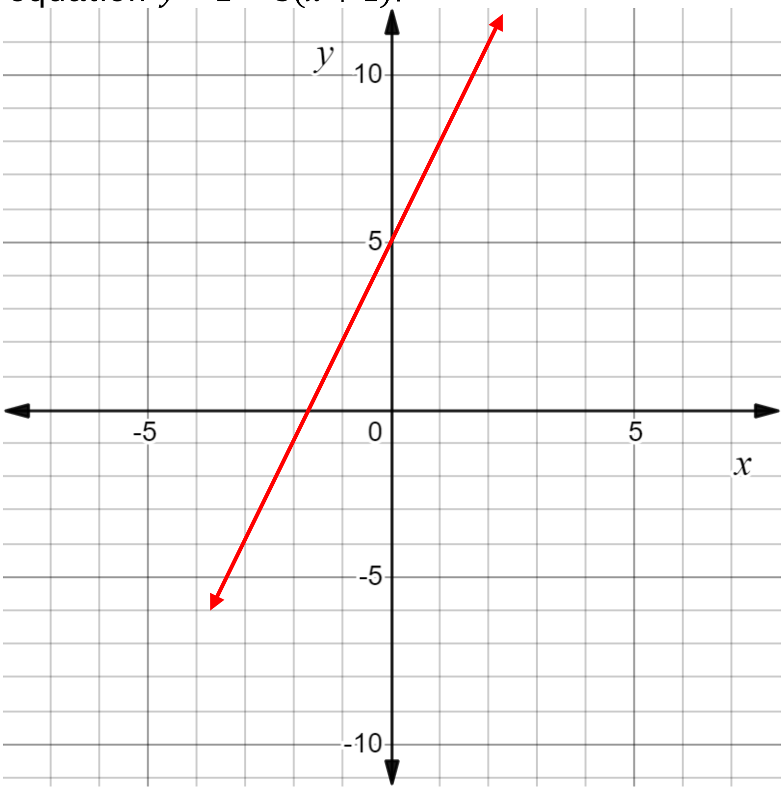
5. Complete the statement about the line $y = -\frac{5}{3}x + 4$.

As x-values increase, y-values __decrease___

6. Complete the table of values for $y - 2 = 3(x+1)$.

x	-2	-1	0	1	2
y	-1	2	5	8	11

7. Graph the equation $y - 2 = 3(x+1)$.



8. Use set-builder notation to complete the table for the line $y - 2 = 3(x+1)$.

Domain	**Range**
$\{x: -\infty < x < \infty\}$	$\{y: -\infty < y < \infty\}$

9. What is the x-intercept of the line $y - 2 = 3(x+1)$?

$\left(-\frac{5}{3}, 0\right)$

10. Complete the statement about the line $y - 2 = 3(x+1)$.

As x-values increase, y-values __increase__

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Ava is hiking a section of the Citrus Hiking Trail in Withlacoochee State Forest. Ava packs trail mix and grapes for snacks. The trail mix provides 5 calories per ounce and the grapes provide 60 calories. The function $C(x) = 5x + 60$ where x is the number of ounces of trail mix Ava eats and C is the total number of calories she consumes. Ava does not want to consume more than 115 calories.

1. Solve $C(x) = 115$.

$115 = 5x + 60$
$-60 \quad\quad -60$
$\frac{55}{5} = \frac{5x}{5}$
$x = 11$

2. Write the range of the function $C(x) = 5x + 60$ using inequalities.

$0 \leq y \leq 115$

3. Determine the x-intercept.

$0 = 5x + 60$
$\frac{-60}{5} = \frac{5x}{5}$
$x = -12$

$(-12,0)$

4. Interpret the x-intercept.

Since x represents the number of ounces of trail mix, it must be a positive number. In this scenario, the x-intercept would tell us she would need to eat -12 ounces of trail mix to consume 0 calories. It is not a viable answer in the context of this scenario.

5. Write the domain of the function $C(x) = 5x + 60$ using inequalities.

$0 \leq x \leq 11$

6. Determine the y-intercept.

$(0,60)$

7. Interpret the y-intercept.

If Ava eats 0 ounces of trail mix, she will consume 60 calories from the grapes.

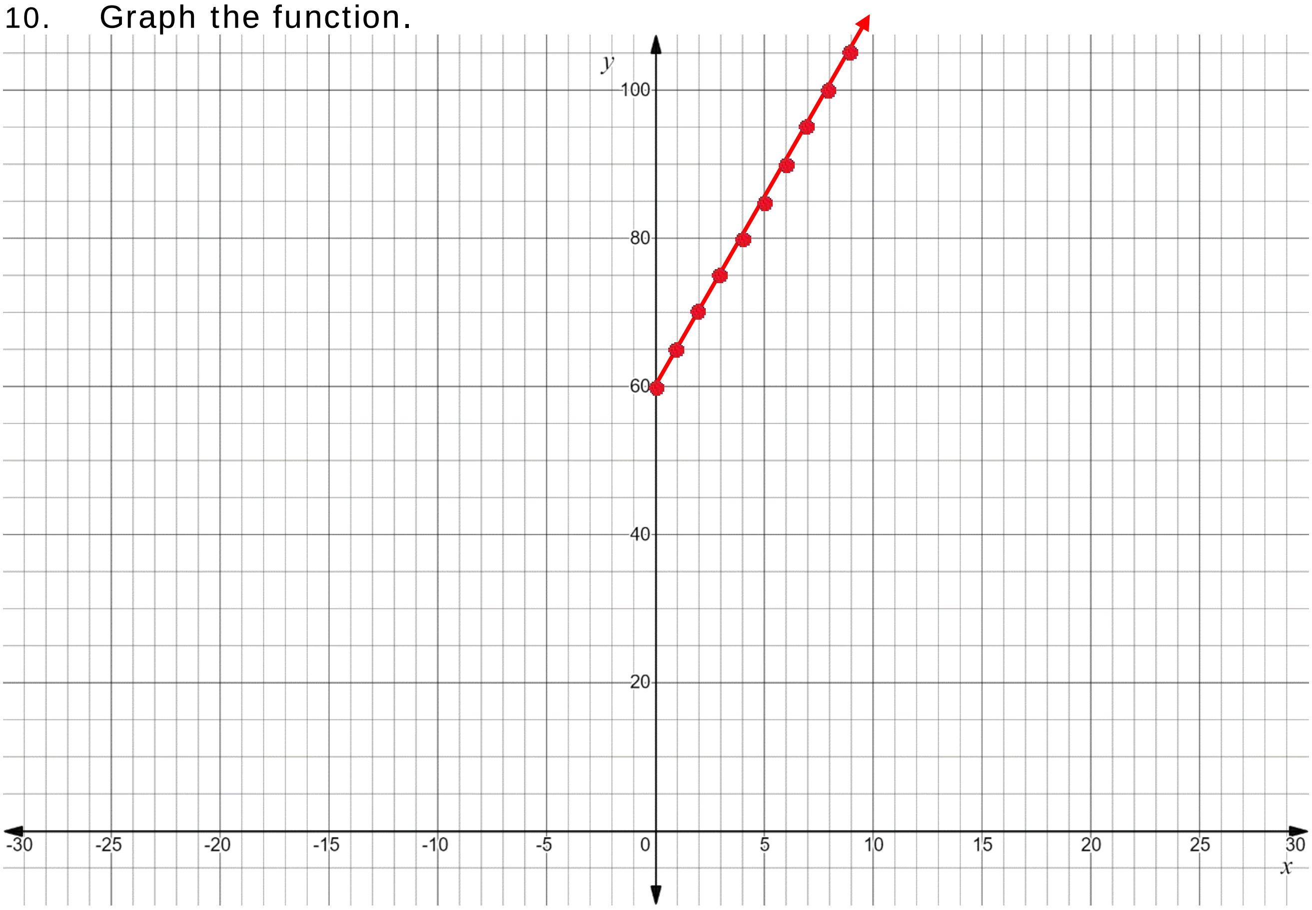
8. Determine the slope.

5

9. Interpret the slope.

Ava consumes 5 calories per ounce of trail mix.

10. Graph the function.



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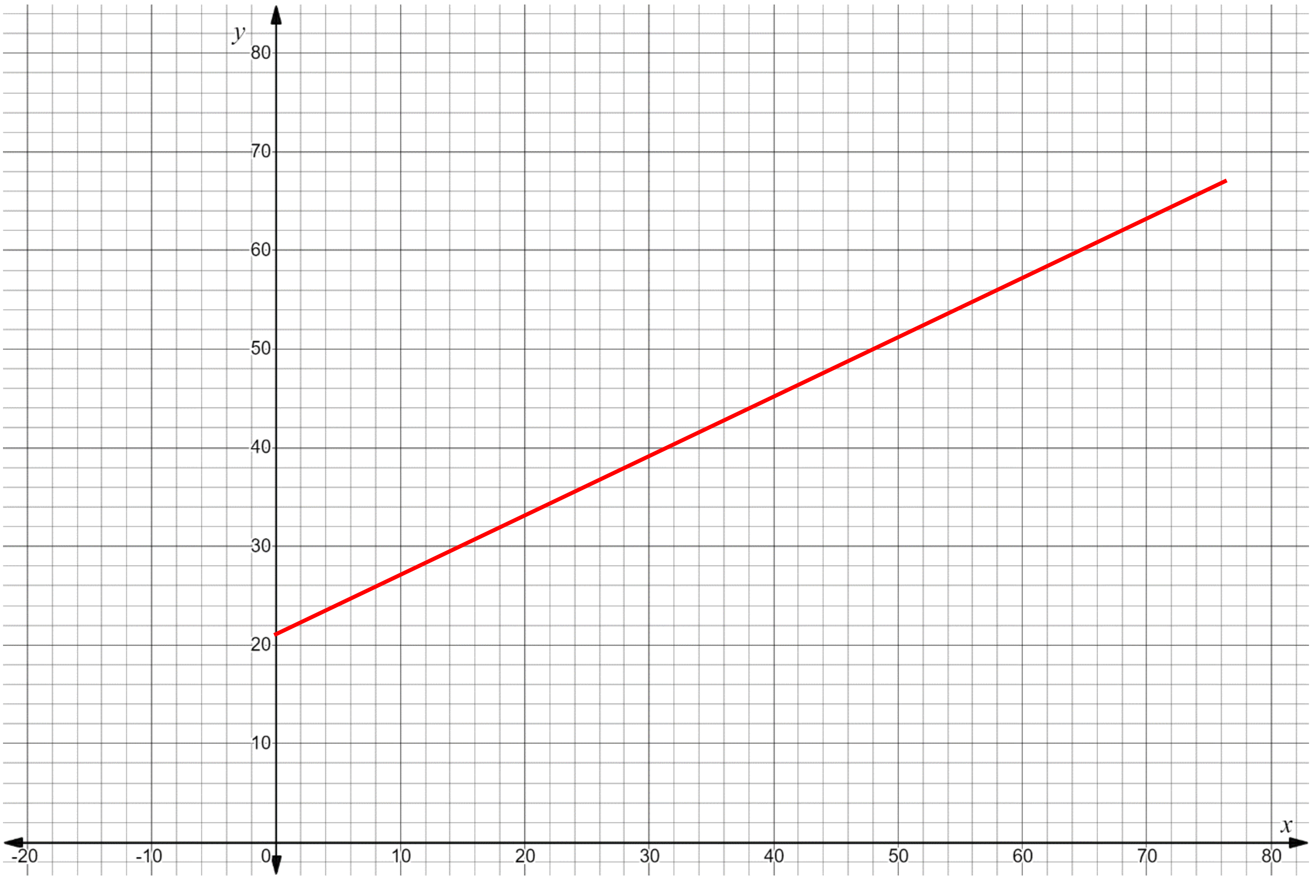
Lesson 5 Homework Period__________________

The table of values shows the linear relationship between x, the number of years since 1900 for all years between 1900 and 2000, and y, the percentage of people in the world who can read.

x	**Number of years since 1900**	0	10	32	48	73
y	**Percentage of people in the world who can read**	21	27	40.2	49.8	64.8

1. Graph the linear relationship.



2. Place a check in the box of a key feature that you used to help you graph the relationship.

Domain	**Range**	**x-intercept**	**y-intercept**	**Slope**	**Increasing / Decreasing**
✓	✓		✓	✓	✓

3. The equation $-6x + 10y = 210$ models the linear relationship. Place a check in the box of a key feature that can be determined using the equation.

Domain	Range	x-intercept	y-intercept	Slope	Increasing / Decreasing
✓	✓	✓	✓	✓	✓

4. Write the equation $-6x + 10y = 210$ in slope-intercept form.

$\frac{10y}{10} = \frac{6x}{10} + \frac{210}{10}$ $\boxed{y = \frac{3}{5}x + 21}$

5. What domain makes sense for the context?

$\{x|0 \leq x \leq 100\}$

6. What range makes sense for the context?

$\frac{3}{5}\left(\frac{\cancel{100}^{20}}{1}\right) + 21$ $\boxed{\{y|21 \leq y \leq 81\}}$

$60 + 21 = 81$

7. Determine the x –intercept.

$0 = \frac{3}{5}x + 21$

$\frac{5}{\cancel{3}_1} \cdot \frac{-\cancel{21}^{7}}{1} = \frac{5}{3} \times \frac{3}{5}x$

$-35 = x$ $\boxed{(-35, 0)}$

8. Interpret the x –intercept.

35 years before 1900, 0% of the people in the world could read. Not within constraints of this scenario.

9. Determine the slope.

$\frac{3}{5}$

10. Interpret the slope.

There is a 3% increase of the people in the world who can read every 5 years.

Algebra 1
Unit 4
Lesson 6 Homework

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For each real-world representation of a context write an inequality that represents the constraint.

1. A child has to be between 30 and 48 inches tall to ride a kiddie ride.
$30 < x < 48$

2. A family has a budget of $\$400$ to spend on groceries each month.
$0 \leq x \leq 400$

3. A person has to be at least 16 years old to drive alone in the state of Florida.
$x \geq 16$

The function $H(x) = 0.6x + 3.8$ models the height, in inches, of a stack of x cups.

4. Find the value of x for which $H(x) = 10.4$.
$10.4 = 0.6x + 3.8$

$-3.8 \qquad\qquad - 3.8$

$\frac{6.6}{0.6} = \frac{0.6x}{0.6}$

$x = 11$

5. In terms of the context, what does the value of x that is the solution to $H(x) = 10.4$ mean?
A stack of 11 cups would yield a height of 10.4 inches.

6. Find the x-intercept.
$0 = 0.6x + 3.8$

$-\frac{3.8}{0.6} = \frac{0.6x}{0.6}$

$x = -6.\overline{3}$

$(-6.33,0)$

7. Determine if the x-intercept is viable. Explain.
The x-intercept is not viable because you can't have a negative number of cups.

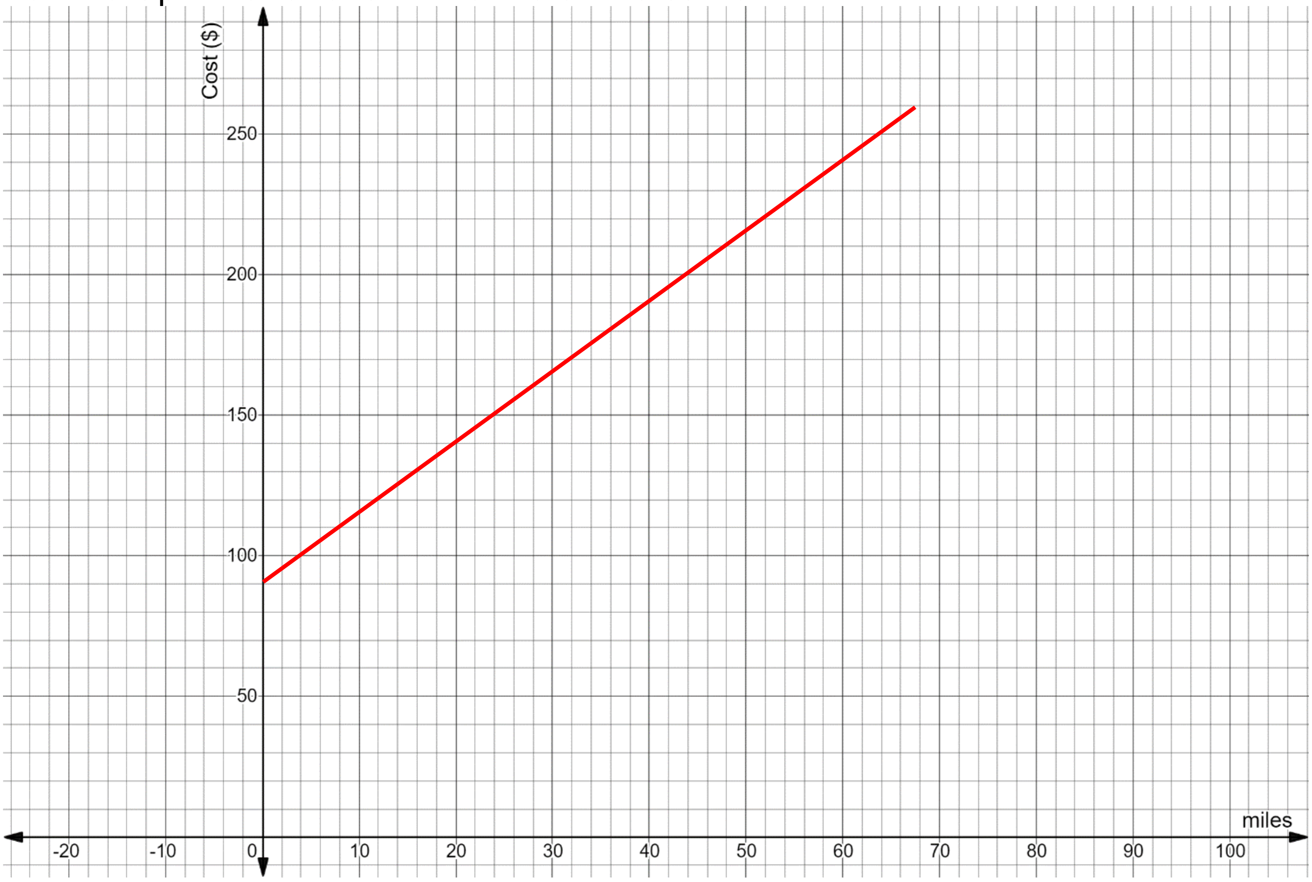
8. What key feature is $H(0)$?
The y-intercept

9. The Peace River State Designated Paddle Trail is 67 miles long. Julia is planning on renting a canoe for a canoe trip on the Peace River Paddle Trail. The canoe outfitter charges $\$90$ to transport the canoe and supplies to the access point. The outfitter charges $\$2.50$ for each mile of the trail that Julia plans to travel. To plan her budget Julia used the function $C(m) = 2.5m + 90$ where C is the cost in dollars and m is the number of miles she travels.

Select all of the constraints for the context.

- [x] $m \geq 0$
- [x] $m \leq 67$
- [ ] $m \leq 90$
- [ ] $C \geq 0$
- [x] $C \geq 90$
- [ ] $C \leq 257.5$

10. Graph the function in terms of the context.



Algebra 1
Unit 4
Lesson 7 Homework

Name Answer Key ___________
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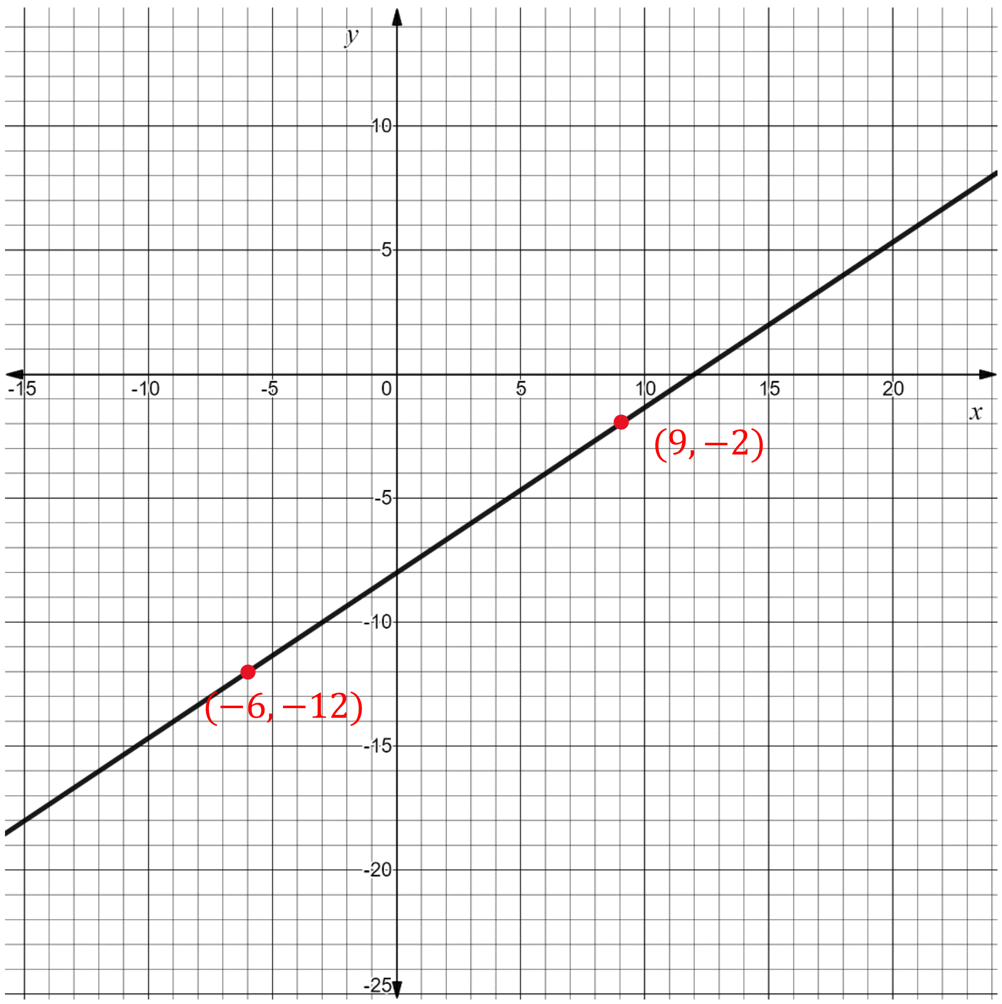
The function $f(x) = \frac{2}{3}x - 8$ is shown on the coordinate grid.



1. Use the graph to find the solution $f(x) = -2$. Label the point that represents the solution.
9

2. Solve $\frac{2}{3}x - 8 = -2$. Show your work.
$+8 \quad +8$
$\cancel{\frac{3}{2}} \cdot \cancel{\frac{2}{3}}x = \frac{\cancel{6}^{3}}{1} \cdot \frac{3}{\cancel{2}_{1}}$
$\boxed{x = 9}$

3. Use the graph to find the solution $f(x) = -12$. Label the point that represents the solution.
$(-6, -12)$

4. Solve $\frac{2}{3}x - 8 = -12$. Show your work.
$+8 \quad +8$
$\cancel{\frac{3}{2}} \cdot \cancel{\frac{2}{3}}x = \frac{\cancel{-4}^{-2}}{1} \cdot \frac{3}{\cancel{2}_{1}}$
$\boxed{x = -6}$

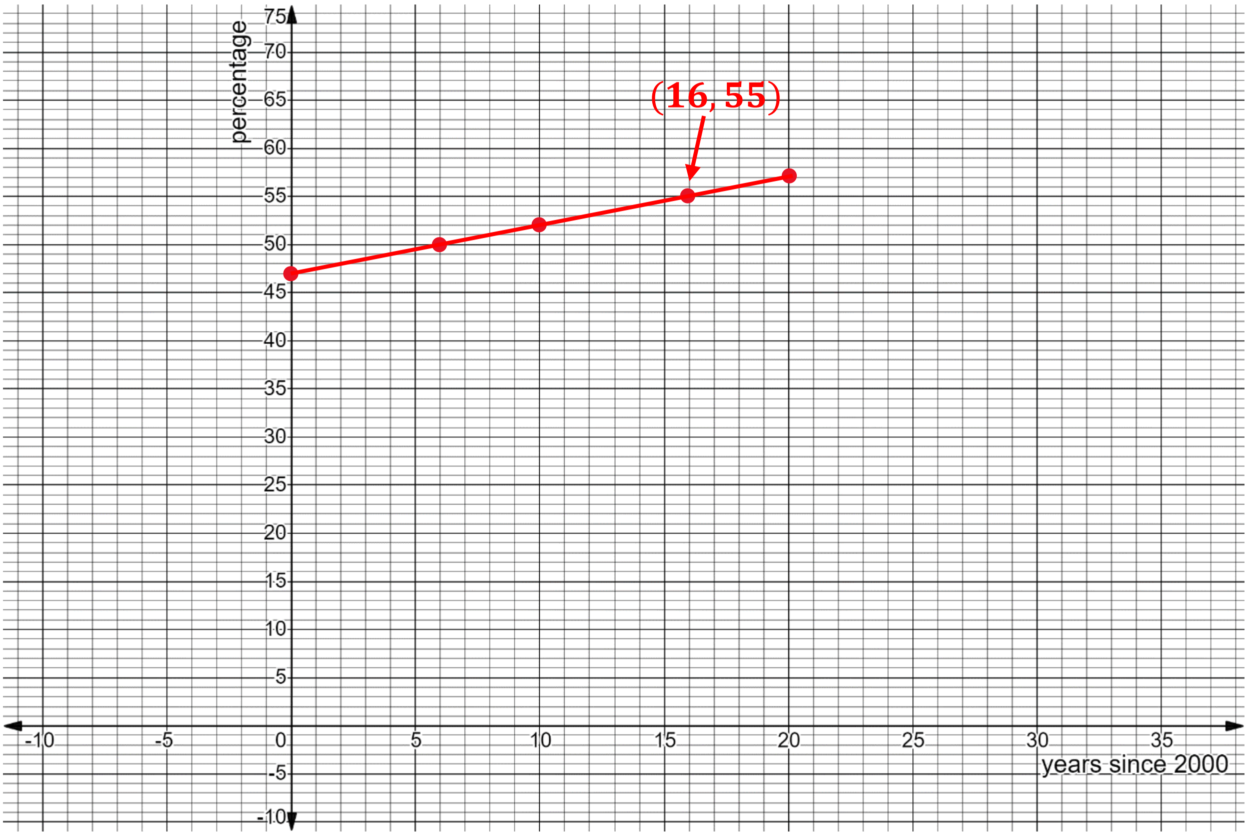
From 2000 to 2020, a world health organization collected data on the percentage of a country's population that had access to safe drinking water. The function $W(t) = 0.5t + 47$ models the percentage of the population in Guatemala that has safe drinking water where t is the number of years since 2000 and W is the percentage of the population that has access.

5. Explain why the values of t should have a constraint.
   The values of t should have a constraint because the data was only measured for 20 years (2000 to 2020).

6. Write the constraint for the values of t using set builder notation.
   $\{x|0 \leq x \leq 20\}$

7. Write the constraint for the values of W using set builder notation.
   $\{w|47 \leq w \leq 57\}$

8. Graph the function using the constraints from questions 6 and 7.



9. Use the graph to find the solution $W(t) = 55$. Label the point that represents the solution.
   16

10. Solve $0.5t + 47 = 55$. Show your work.

$$-47 \quad -47$$

$$\frac{0.5t}{0.5} = \frac{8}{0.5}$$

$$t = 16$$

Algebra 1
Unit 4
Lesson 8 Homework

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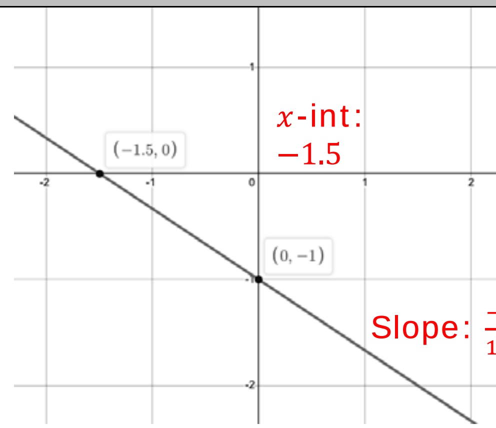
For questions 1 through 4, use the table showing Lines A and B.

Line A	**Line B**
x-int: $0 = \frac{-2}{3}x + 3$ $\frac{-3}{2} \cdot \frac{-3}{1} = \frac{-3}{2} \cdot \frac{-2}{3}x$ $\frac{9}{2} = 4.5 = x$ $B(x) = -\frac{2}{3}x + 3$ Slope: $-\frac{2}{3}$	 x-int: -1.5 Slope: $\frac{-1}{1.5} = -\frac{2}{3}$

1. Which linear function has the larger slope?
The slopes are the same.

2. Which line has the larger x-intercept?
Line A

3. Name all the key feature(s) that are the same for both linear functions.
Both linear functions have the same slope, domain, and range. They are also both decreasing.

4. Create a linear function that has the same y-intercept as Line A.
Answers may vary. $C(x) = \frac{1}{4}x + 3$

5. Complete the statements below to describe the end behavior of Line A.

a. As $x \to \infty$, $y \to$
- ✔ $-\infty$
- o ∞

b. As $x \to -\infty$, $y \to$
- o $-\infty$
- ✔ ∞

For questions 6 through 10, use the table showing Lines A and B.

<table>
<tr><th>Line A</th><th>Line B</th></tr>
<tr><td>x-int: $0 = 2x + 6$
$-6 \quad\quad -6$
$-3 = x$
y-int: 7
A line passes through the points $(1, 9)$ and $(4, 15)$.
Slope: $\frac{15-9}{4-1} = \frac{6}{3} = 2$
$y - 9 = 2(x - 1)$
$y - 9 = 2x - 2$
$+9 \quad\quad +9$
$y = 2x + 7$</td>
<td>x-int: -15 Slope: $\frac{7-6}{6-3} = \frac{1}{3}$
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-3</td><td>4</td></tr>
<tr><td>0</td><td>5</td></tr>
<tr><td>3</td><td>6</td></tr>
<tr><td>6</td><td>7</td></tr>
</table>
y-int: 5</td></tr>
</table>

$y - 7 = \frac{1}{3}(x - 6)$

$y - 7 = \frac{1}{3}x - 2$

$+7 \quad\quad +7$

$y = \frac{1}{3}x + 5$

6. Which linear function has the smaller slope?
Line B

7. Which linear function has the larger y-intercept?
Line A

8. Name the key feature(s) that are different for the two functions.
Line A and Line B have different x- and y-intercepts and slopes.

9. Create a linear function that has the same slope as Line B.
Answers may vary. $D(x) = \frac{1}{3}x - 6$

10. Consider the end behavior for the two functions.

a. Complete the statement.

Lines A and B have
- ✓ the same end behavior.
- o different end behaviors.

b. Write the end behavior of the functions.
As $x \to \infty$, $y \to \infty$.
As $x \to -\infty$, $y \to -\infty$.

Algebra 1
Unit 4
Lesson 9 Homework

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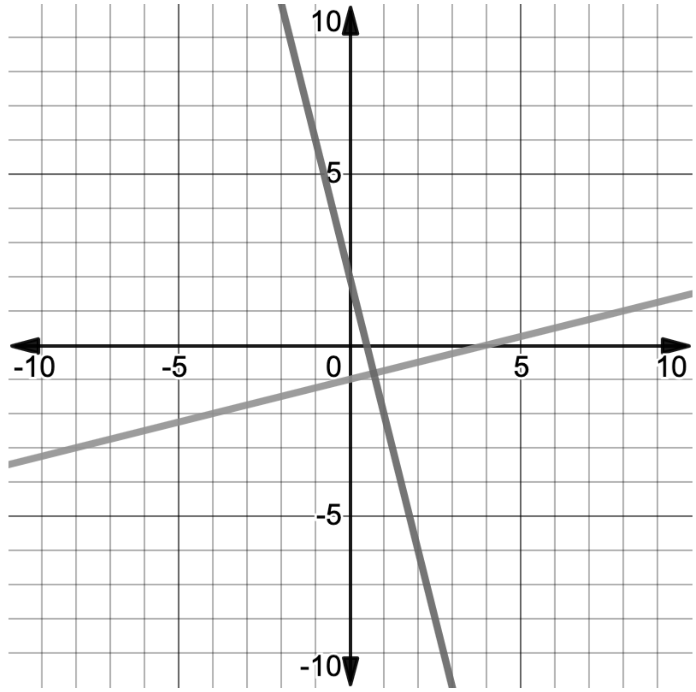
Use the following information to answer questions 1 through 5.

The lines of the equations $y - 6 = -4(x + 1)$ and $x - 4y = 4$ are shown on the same coordinate grid.



1. What is the slope of the line representing the equation $y - 6 = -4(x + 1)$?

$y - 6 = -4x - 4$

$y = -4x - 4 + 6$

$y = -4x + 2$

$m = -4$

2. What is the slope of the line representing the equation $x - 4y = 4$?

$\frac{x}{4} - \frac{4}{4} = \frac{4y}{4}$

$\frac{x}{4} - 1 = y$

$m = \frac{1}{4}$

3. What is the slope of a line that is parallel to $x - 4y = 4$?

$m = \frac{1}{4}$

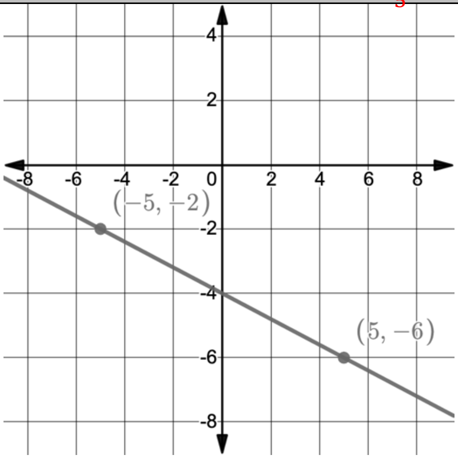
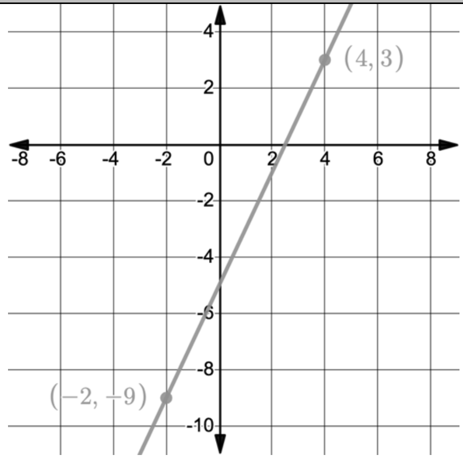
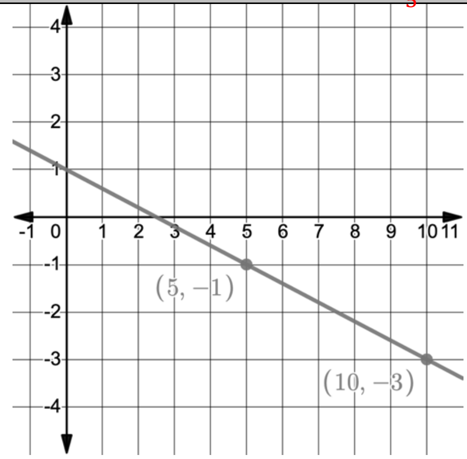
4. Explain how to determine if two lines are parallel.

They have the same slopes but different y-intercepts.

5. Determine if the lines are parallel, perpendicular, or neither.

These lines are perpendicular.

Use the following lines to answer questions 6 through 8.

<table>
<tr><th>Line A $y=-\frac{2}{5}x-4$</th><th>Line C $y=2x-5$</th><th>Line D $y=-\frac{2}{5}x+1$</th></tr>
<tr><td>
</td><td>
</td><td>
</td></tr>
<tr><th>Line E</th><th>Line G</th><th>Line Y</th></tr>
<tr><td>$y=-0.5x+3$</td><td>$y-4=-(x+5)$</td><td>$-5y=2x+5$</td></tr>
</table>

$y=-x-1$ $\qquad$ $y=-\frac{2}{5}x-1$

6. Which lines are parallel?

   A, D, & Y

7. Which lines are perpendicular?

   C & E

8. What is the slope of a line that is perpendicular to Line G?

   1

9. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation K	Equation V
$7x-4y=8$	$4x-7y=14$

$y=\frac{7}{4}x-2$ $\qquad$ $y=\frac{4}{7}x-2$

Neither

10. Determine if the lines of the two equations are parallel, perpendicular, or neither.

Equation M	Equation R
$3x-2y=6$	$y+8=\frac{3}{2}(x+6)$

$y=\frac{3}{2}x-3$ $\qquad$ $y=\frac{3}{2}x+1$

Parallel

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1. Write the equation of the line that passes through $(4, 2)$ and is parallel to $y = 3x + 1$ in point-slope form.
$y - 2 = 3(x - 4)$

2. Write the equation of the line that is parallel to $y = -\frac{3}{2}$ and passes through point $(2, 3)$.
$y = 3$

3. Write the equation of the line that contains point $(0, 5)$ and is parallel to $2x + 3y = 6$ in slope-intercept form.
$2x + 3y = 6 - 2x$
$-2x$
$\frac{3y}{3} = \frac{-2x+6}{3}$
$y = \frac{-2}{3}x + 2$

$5 = \frac{-2}{3}(0) + b$
$5 = b$

$y = -\frac{2}{3}x + 5$

4. Write the equation of the line that is parallel to $x - 2 = 0$ and passes through point $(1, -2)$.
$x = 1$

5. Write the equation of the line that passes through the origin and is parallel to $y + 9 = -(x - 2)$ in standard form.
$y - 0 = -1(x - 0)$
$y = -x$
$x + y = 0$

6. Write the equation of the line, in all three forms, that passes through point $(-10, -3)$ and is parallel to $y + \frac{1}{4} = \frac{1}{4}(x - 1)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - (-3) = \frac{1}{4}(x - (-10))$ $y + 3 = \frac{1}{4}(x + 10)$	$-3 = (-10)\left(\frac{1}{4}\right) + b$ $-3 = -\frac{10}{4} + b$ $+\frac{10}{4} \quad +\frac{10}{4}$ $-\frac{1}{2} = b$ $y = \frac{1}{4}x - \frac{1}{2}$	$\left(y + 3 = \frac{1}{4}x + \frac{10}{4}\right)4$ $4y + 12 = x + 10$ $-x \; -12 \quad -x \; -12$ $-x + 4y = -2$ *or* $x - 4y = 2$

7. Write the equation of a line that is parallel to $y = x$ and contains point $(-2, 2)$.

$2 = 1(-2) + b$

$2 = -2 + b$

$+2 \quad +2$

$4 = b$

$y = x + 4$

8. Line N passes through $\left(\frac{1}{2}, -6\right)$ and is parallel Line Q which has an undefined slope. Write the equation of Line N.

$x = \frac{1}{2}$

9. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(0, 5)$ and $\left(\frac{5}{6}, 0\right)$ and passes through point $(4, 4)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-5}{\frac{5}{6}-0} = \frac{-5}{\frac{5}{6}} = -6$ $y - 4 = -6(x - 4)$	$4 = 4(-6) + b$ $4 = -24 + b$ $+24 \quad +24$ $28 = b$ $y = -6x + 28$	$y - 4 = -6x + 24$ $+6x \quad +4 \quad +6x \quad +4$ $6x + y = 28$

10. Write the equation of the line, in all three forms, that is parallel to the line that passes through the points $(-6, 1)$ and $(10, 0)$ and passes through point $(-7, -2)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-1}{10+(+6)} = \frac{-1}{16}$ $y - (-2) = \frac{-1}{16}(x - (-7))$ $y + 2 = \frac{-1}{16}(x + 7)$	$-2 = (-7)\left(-\frac{1}{16}\right) + b$ $-2 = \frac{7}{16} + b$ $\frac{-7}{16} \quad \frac{-7}{16}$ $-\frac{39}{16} = b$ $y = \frac{-1}{16}x - \frac{39}{16}$	$y + 2 = \frac{-1}{16}x - \frac{7}{16} - 2$ -2 $\left(y = \frac{-1}{16x} - \frac{39}{16}\right)16$ $16y = -x - 39$ $+x \quad +x$ $x + 16y = -39$

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1. Write the equation of the line that passes through $(-2, 1)$ and is perpendicular to $y = 2x - 3$ in point-slope form. Slope $= -\frac{1}{2}$

$y - 1 = \frac{-1}{2}(x - -2)$

$y - 1 = -\frac{1}{2}(x + 2)$

2. Write the equation in slope-intercept form of the line that is perpendicular to $x = 4$ and passes through point $(4, -5)$.

$y = -5$

3. Write the equation of the line that contains point $(-2, 2)$ and is perpendicular to $x - 2y = 5$ in standard form. Slope $= -2$

$-x \quad -x$

$\frac{-2y}{-2} = \frac{5-x}{-2}$

$y = \frac{1}{2}x - \frac{5}{2}$

$y - 2 = -2(x + 2)$

$y - 2 = -2x - 4$

$+2x + 2 \quad +2x + 2$

$2x + y = -2$

4. Write the equation of the line that is perpendicular to $y + 2 = 0$ and passes through point $(3, -3)$.

$x = 3$

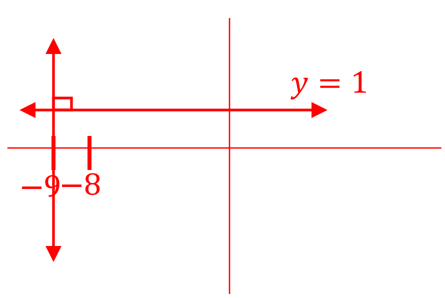
5. Line B passes through $(11, 13)$ and is perpendicular Line C which has an undefined slope. What is the slope of line B?

$m = 0$

6. Write the equation of the line that is perpendicular to the line that passes through the points $(0, 1)$ and $(-8, 1)$ and passes through point $(-9, 12)$.

$\frac{1-1}{-8-0} = \frac{0}{-8} = 0 =$ slope

Equation of line perpendicular → $x = -9$



$y - 1 = 0(x - 0)$

$\perp$ slope → undefined $\frac{8}{0}$

7. Write the equation of a line that is perpendicular to $y = x$ and contains point $(-6, 6)$. Slope $= -1$

$6 = (-1)(-6) + b$

$6 = 6 + b$

$-6 \quad -6$

$b = 0$

$y = -x$

8. Write the equation of the line, in all three forms, that passes through point $(5, 15)$ and is perpendicular to $y + 6 = \frac{5}{8}(x - 24)$. Slope $= -\frac{8}{5}$

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 15 = -\frac{8}{5}(x - 5)$	$15 = 5\left(\frac{-8}{5}\right) + b$ $15 = -8 + b$ $\underline{+8 \quad +8}$ $23 = b$ $y = -\frac{8}{5}x + 23$	$y - 15 = \frac{-8}{5}x + 8$ $+\frac{-8}{5}x + 15$ $+\frac{-8}{5}x + 15$ $\left(\frac{8}{5}x + y = 23\right)5$ $8x + 5y = 115$

9. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(0, -9)$ and $(54, 0)$ and passes through point $(1, 9)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{0-(-9)}{54-0} = \frac{9}{54} \rightarrow \frac{-54}{9} = -6$ $y - 9 = -6(x - 1)$	$9 = -6(1) + b$ $9 = -6 + b$ $\underline{+6 \quad +6}$ $15 = b$ $y = -6x + 15$	$y - 9 = -6x + 6$ $+6x + 9 \quad +6x + 9$ $6x + y = 15$

10. Write the equation of the line, in all three forms, that is perpendicular to the line that passes through the points $(-14, 22)$ and $(2, -2)$ and passes through point $(0, 4)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$\frac{-2-22}{2+14} = -\frac{24}{16} \rightarrow \frac{16}{24} = \frac{2}{3}$ $y - 4 = \frac{2}{3}(x - 0)$	$4 = 0\left(\frac{2}{3}\right) + b$ $4 = b$ $y = \frac{2}{3}x + 4$	$\left(y - 4 = \frac{2}{3}x\right)3$ $3y - 12 = 2x$ $-2x + 12 \quad -2x + 12$ $-2x + 3y = 12$ or $2x - 3y = -12$

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1. Write the equation of the line that passes through $(7,3)$ and is parallel to $4x+2y=10$.

$-4x \qquad -4x$

$\frac{2y}{2}=\frac{10-4x}{2}$

$y=-2x+5$

slope

$3=(-2)(7)+b$

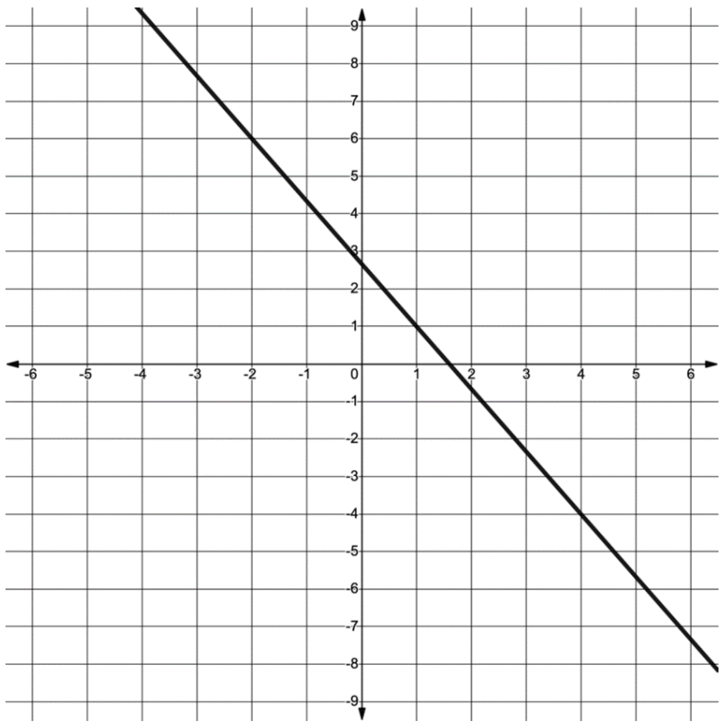
$3=-14+b$

$+14 \quad +14$

$17=b$

$y=-2x+17$

Use the line shown on the coordinate grid to complete questions 2 through 4.



2. What is the slope of the line that is parallel to the line shown on the coordinate grid?

$-\frac{5}{3}$

3. Write the equation of the line that is parallel to the line shown on the coordinate grid and that passes through $(6,-4)$.

$-4=\frac{-5}{3}(6)+b$

$-4=-10+b$

$+10 \quad +10$

$b=6$

$y=-\frac{5}{3}x+6$

4. Write the equation of the line that is perpendicular to the line shown on the coordinate grid and that passes through $(6, -4)$.

$-4 = \frac{3}{5}(6) + b$

$-4 = \frac{18}{5} + b$

$-\frac{18}{5} \quad -\frac{18}{5}$

$b = -7\frac{3}{5}$

$y = \frac{3}{5}x - 7\frac{3}{5}$

5. Write the equation of the line in standard form that is perpendicular to $y = 2x + 5$ and passes through $(-2, 3)$.

slope

$3 = -2\left(-\frac{1}{2}\right) + b$

$3 = 1 + b$

$\underline{-1 \quad -1}$

$2 = b$

$y = -\frac{1}{2}x + 2$

$+\frac{1}{2}x \quad +\frac{1}{2}x$

$2\left(\frac{1}{2}x + y = 2\right)$

$x + 2y = 4$

Use the following information about Line P to complete questions 6 through 8.

Line P passes through $(-23, 33)$ and $\left(\frac{7}{4}, 0\right)$. $\frac{0-33}{\frac{7}{4}-(-23)} = \frac{-33}{\frac{99}{4}} = \frac{-33 \cdot 4}{99} = -\frac{4}{3}$ slope

6. Line W passes through $(0, 9)$ and is parallel to Line P. Write the equation of Line W in slope-intercept form.

$9 = \frac{-4}{3}(0) + b$

$9 = b$

$y = -\frac{4}{3}x + 9$

7. Line T is perpendicular to Line P. What is the slope of Line T?

$\frac{3}{4}$

8. Line T passes through $(-5, 2)$ Write the equation of Line T in all three forms.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y - 2 = \frac{3}{4}(x + 5)$	$2 = -5\left(\frac{3}{4}\right) + b$ $2 = \frac{-15}{4} + b$ $+\frac{15}{4} \quad +\frac{15}{4}$ $\frac{23}{4} = b$ $y = \frac{3}{4}x + \frac{23}{4}$	$\left(y = \frac{3}{4}x + \frac{23}{4}\right)4$ $4y = 3x + 23$ $\underline{-3x \quad -3x}$ $-3x + 4y = 23$ *or* 3x – 4y = -23

9. Write the equation of the line that is parallel to $9x + y = -2$ and passes through point $(2, -26)$ in slope-intercept form.

$-9x \quad -9x$

$y = -9x - 2$

$-26 = 2(-9) + b$

$-26 = -18 + b$

$+18 \quad +18$

$-8 = b$

$y = -9x - 8$

10. Write the equation of the line that is perpendicular to $\underline{x} = -\underline{y}$ and passes through point $\left(\frac{7}{2}, \frac{7}{2}\right)$.

$-1 \quad -1$

$y = -x$

$\frac{7}{2} = (+1)\left(\frac{7}{2}\right) + b$

$\frac{7}{2} = \frac{7}{2} + b$

$-\frac{7}{2} \quad -\frac{7}{2}$

$0 = b$

$y = x$